

challenges, especially when that product needs to communicate wirelessly in an unpredictable environment. From software updates to RF interference, and from interoperability with phones to user privacy, product designers must operate with a wide-angle view.

Thinking beyond the hardware

At the software and algorithmic levels, modern wireless standards, such as Wi-Fi, Bluetooth, Zigbee, and Thread, include mechanisms like carrier sensing, listen-before-talk, and frequency hopping to mitigate interference and reduce packet collisions. However, it is crucial for product designers to integrate these technologies effectively from the start.

Shielding solutions are commonly used to minimise external RF noise, though completely eliminating it is not feasible. These design choices must be made proactively across physical, protocol, and application layers.

Interoperability is essential but challenging for IoT developers, as devices must connect with ecosystems such as Apple HomeKit, Google Home, or Amazon Alexa. Mobile apps for configuration, updates, and user interaction are also necessary. With numerous hardware and OS variations and no control over other devices, testing and maintenance can become difficult. Even devices following the same BLE specifications may behave inconsistently across different phones, and changes like OS updates can break functionality, damaging user trust.

Security is a significant concern, as a compromised device in the network can expose others, especially if it connects to a hub that is also linked to vulnerable devices. Data privacy is another risk; if not properly encrypted, sensitive

information on a mobile app could be accessible to other apps, particularly in Android ecosystems where app permission settings vary.

The standards gap and design accountability

While compliance testing exists and catches major violations, it often fails to address subtle, performance-impacting issues such as RF noise susceptibility or inadequate shielding. For example, many poorly designed LED bulbs generate significant RF noise in 300-400MHz band, causing garage door openers operating in this band to stop functioning. This is not a combination that is generally tested in labs, but rather one of the many practical conditions that are encountered when the product is in use.

This raises an important question: Should we push for higher standards, or should we over-engineer our designs to tolerate poorly behaved neighbours?

The answer is ‘both.’ Practically, the timelines for product development are very tight in a competitive market, and standardisation usually takes a longer time. If an issue is real and present, address it in your design now through better shielding, smarter firmware, or more rigorous testing.

Physical deployment environments present another set of hurdles. Consider the performance of a Thread-based smart home system in Europe versus India. In Germany, lightweight construction using wood and drywall allows RF signals to propagate with minimal loss. In contrast, Indian homes with concrete and brick walls significantly attenuate signals. This is a physics problem, not a design flaw.

So, how do we deal with it? By adapting our system architecture. In RF-hostile environments, using a mesh network topology with

additional repeaters can improve coverage and reliability. Smart deployment and planning can mitigate environmental constraints, but the problem must be acknowledged from the beginning.

Building for the ecosystem, not just the product

Ultimately, the success of any connected product depends on its ability to function effectively within a dynamic and inconsistent ecosystem. From software updates to RF co-existence, from security to physical deployment, designers must plan for variability. Customers don’t blame the phone, the bulb or the OS. They blame the product—your product.

That is why it is critical to:

- Conduct extensive interoperability and regression testing
- Anticipate edge cases and rare failure modes
- Design with margins that account for unpredictable environments.
- Stay engaged with standards bodies
- Be proactive, not reactive, in addressing ecosystem challenges.

Transitioning from good technology to a successful product requires careful consideration across multiple layers—physical, protocol, application, and environmental. Standards help, but product thinking must extend beyond just specifications. As product designers and engineers, the onus is on us to deliver reliable, secure, and user-friendly solutions. When we do that right, we build not just a product, but trust. And trust is the foundation of every lasting user experience. **EFY**

This article is based on a tech talk session at IEW 2025, Bengaluru, by Srividya Sundar, Consultant – Architect for Connectivity and Smart Systems, Ather Energy (Individual Tech Consultant). It was transcribed and curated by Akanksha Sondhi Gaur, Senior Technical Journalist at EFY.